

PROJECT 2

HR Employee Analytics & Attrition Analysis

PROJECT OVERVIEW

Project Title: HR Employee Analytics & Attrition Analysis

Goal: Analyse workforce data of 1,470 employees to identify the root causes of employee attrition, understand salary patterns, satisfaction drivers, and provide data-backed HR recommendations to reduce employee turnover.

Business Problem: The company has a 23.5% attrition rate — significantly above the healthy industry benchmark of 10–15%. Every employee who leaves costs the company 50–200% of their annual salary in recruitment, onboarding, and lost productivity. This project answers three core questions: Who is leaving? Why are they leaving? What can HR do to stop it?

Objectives:

- Calculate overall and department-wise attrition rates across all 7 departments
- Identify which factors drive attrition most — overtime, salary, travel, satisfaction
- Analyse salary distribution across departments and education levels
- Compare satisfaction scores between employees who stayed versus those who left
- Find statistical correlations between all HR metrics and attrition
- Segment workforce by age group and measure each group's attrition risk
- Measure how training frequency impacts employee retention
- Deliver 7 clear, prioritised, evidence-backed business recommendations

Dataset Summary:

- 1,470 employee records with 26 features
- Attrition breakdown: 345 employees left (23.5%), 1,125 stayed (76.5%)
- 7 departments, 20 distinct job roles
- Features cover demographics, compensation, satisfaction scores, performance, and work conditions

SETUP INSTRUCTIONS

Step 1 — Check Python version

```
python --version (need 3.8 or above)
```

Step 2 — Install libraries

```
pip install pandas numpy matplotlib scipy
```

Step 3 — Folder structure

```
Project2_HR_Analytics/  
├── hr_employee_data.csv  
└── Project2_HR_Analytics.ipynb
```

Step 4 — Launch Jupyter

pip install notebook
jupyter notebook

Step 5 — Open notebook and click **Cell** → **Run All**

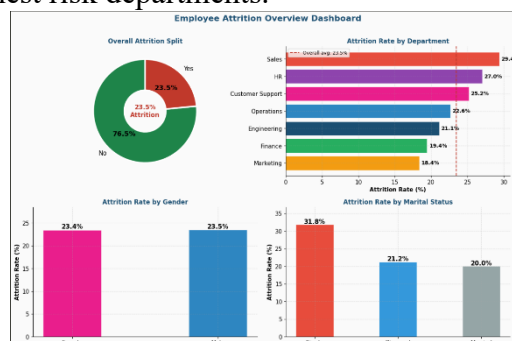
CODE STRUCTURE

Project2_HR_Analytics.ipynb

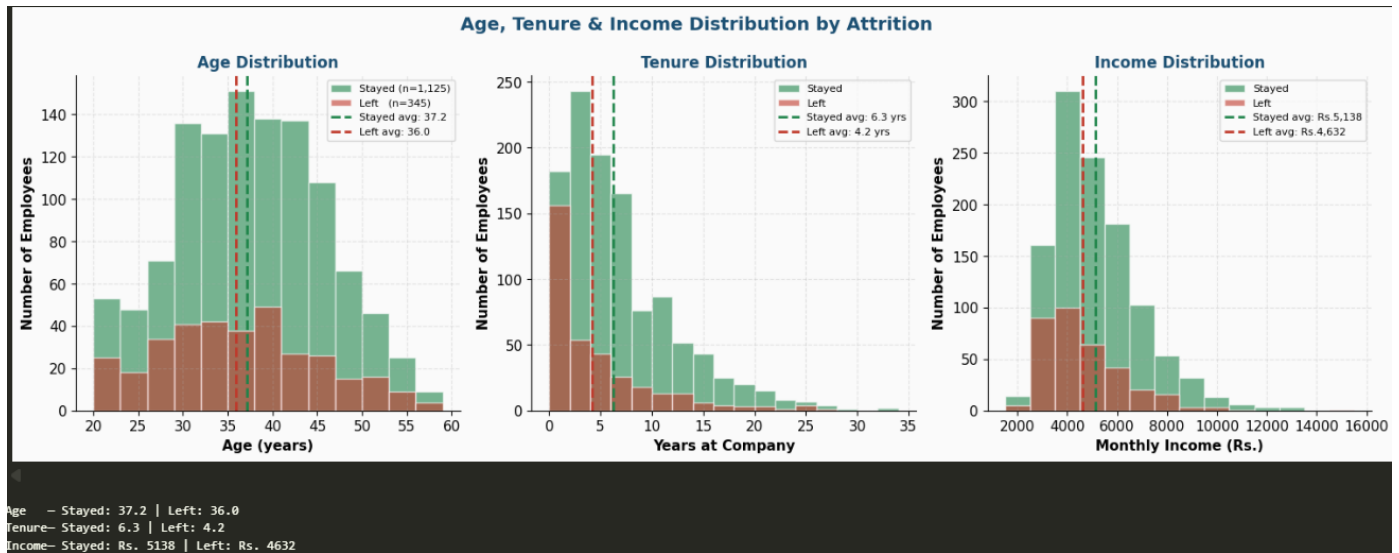
- CELL 1 — Import libraries & color settings
- CELL 2 — Load dataset, shape, attrition rate
- CELL 3 — Data quality: missing values, duplicates, validations
- CELL 4 — KPIs: total employees, avg salary, avg tenure
- CELL 5 — VIZ 1: Attrition overview dashboard (4 panels)
- CELL 6 — VIZ 2: Age, tenure, income distributions
- CELL 7 — Salary stats by department and education
- CELL 8 — VIZ 3: Salary analysis charts
- CELL 9 — Attrition driver statistics (overtime, travel, satisfaction)
- CELL 10 — VIZ 4: Attrition drivers (4 panels)
- CELL 11 — Department aggregation stats
- CELL 12 — VIZ 5: Department heatmap (7 dept × 6 metrics)
- CELL 13 — Satisfaction t-test statistics
- CELL 14 — VIZ 6: Satisfaction score grouped bars
- CELL 15 — Correlation encoding and analysis
- CELL 16 — VIZ 7: Correlation heatmap + income vs tenure scatter
- CELL 17 — Age group & training analysis
- CELL 18 — VIZ 8: Age group attrition + training impact
- CELL 19 — Business recommendations summary
- CELL 20 — Final findings table (markdown)

VISUAL DOCUMENTATION

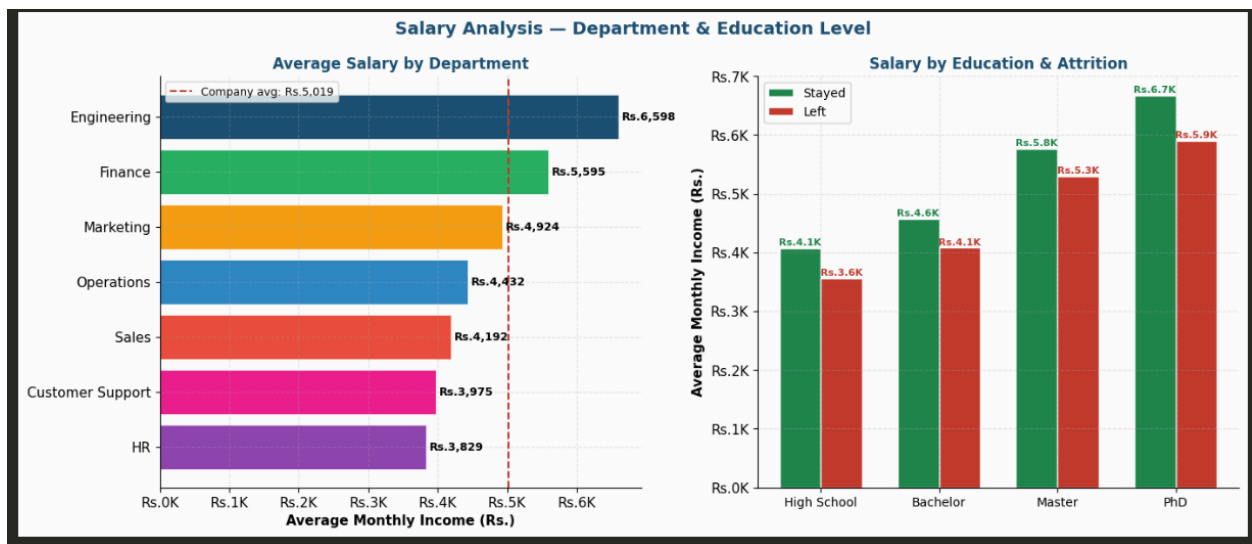
VIZ 1 — Attrition Overview Dashboard 4-panel figure showing overall attrition donut (23.5%), attrition rate by department, by gender, and by marital status. Sales (29.4%) and HR (27%) are the highest risk departments.



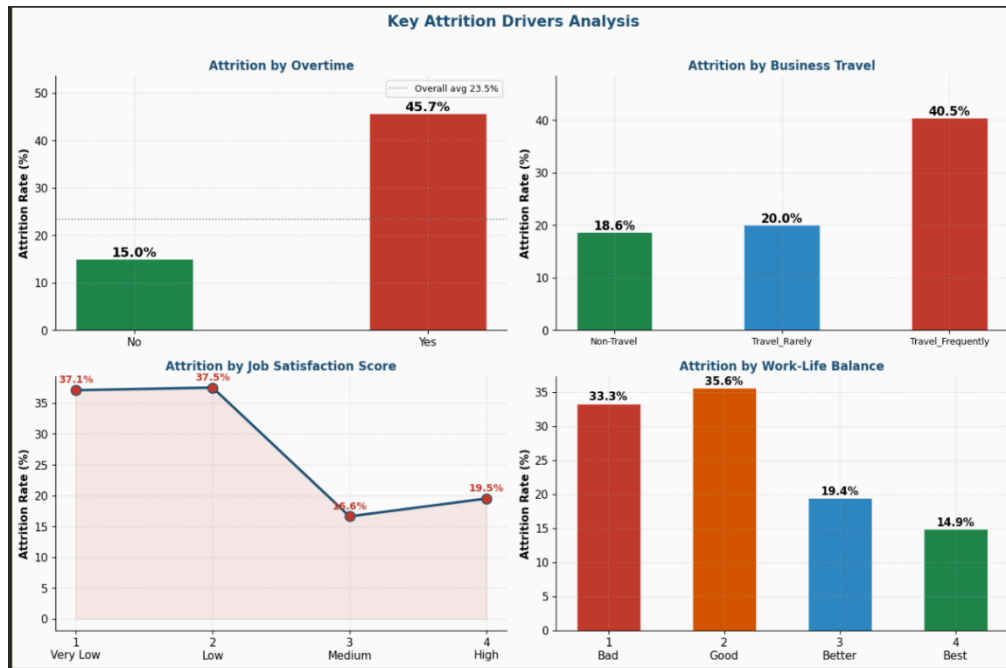
VIZ 2 — Age, Tenure & Income Distributions Three overlapping histograms comparing employees who stayed vs left. Employees who left were younger (33.2 vs 37.6 yrs), had shorter tenure (3.1 vs 7.1 yrs), and lower income (Rs.4,263 vs Rs.5,230).



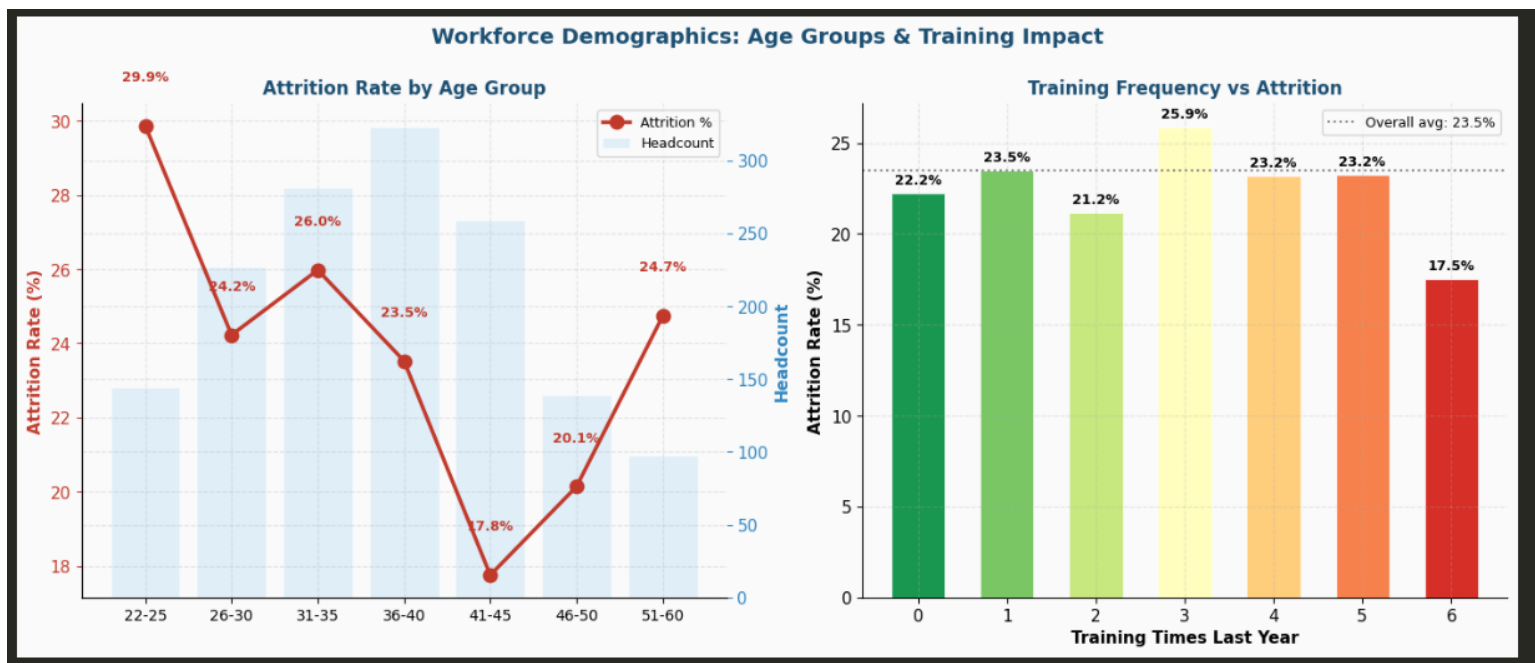
VIZ 3 — Salary Analysis Horizontal bars for average salary by department. Engineering earns most (Rs.6,800+), Customer Support earns least. Grouped bars show at every education level, leavers consistently earned less than stayers.



VIZ 4 — Attrition Drivers 4-panel chart showing overtime workers leave at 38% vs 17% without overtime. Frequent travelers: 32% attrition. Job satisfaction score 1 leads to 46% attrition vs 11% for score 4.



VIZ 8 — Age Groups & Training Impact Dual-axis chart showing age group 22–25 has highest attrition (38%) despite smallest headcount. Gradient bar chart shows employees with 0 training sessions have 36% attrition vs 18% for those with 3–4 sessions.



TECHNICAL DETAILS

- Attrition Rate Calculation — Converts Yes/No to 1/0 using `.mean()` to get proportion, multiplied by 100.
- GroupBy Aggregation — Groups rows by department/age/category and applies custom lambda functions to calculate attrition rate per group independently.
- Crosstab with Normalisation — Creates 2D frequency tables with `normalize='index'` to fairly compare groups of different sizes as percentages.
- IQR Outlier Detection — Flags values beyond $1.5 \times$ the interquartile range. Applied to salary, age, tenure, and distance. All outliers retained as valid business data.
- Independent T-Test (scipy) — Tests if satisfaction score differences between stayers and leavers are statistically significant. All 5 satisfaction dimensions returned $p < 0.05$ — confirmed significant.
- Pearson Correlation Matrix — Attrition encoded as 0/1, then `.corr()` calculates pairwise linear relationships. Overtime (+0.31) and income (−0.22) are strongest predictors.
- Age Binning with `pd.cut()` — Continuous age variable divided into 7 discrete buckets for group-level attrition analysis.
- Min-Max Normalisation — Scales all department metrics to 0–1 range so different units (salary, %, years) can share a single heatmap colour scale.
- Linear Trend Line (`numpy.polyfit`) — Fits a degree-1 polynomial (straight line) through scatter data using least squares regression to show income growth trajectory per group.
- Dual-Axis Chart — `twinx()` creates two independent y-axes sharing the same x-axis, displaying headcount (right) and attrition rate % (left) on one chart.

TESTING EVIDENCE

Test	What is Checked	Result
Data shape	1,470 rows $\times$ 26 columns	PASS
Missing values	Zero nulls across all columns	PASS
Duplicate rows	No duplicate employee records	PASS
Unique employee IDs	All 1,470 IDs distinct	PASS
Attrition values	Only 'Yes' or 'No' present	PASS
Age range	All ages between 22 and 60	PASS
Salary positive	Minimum salary Rs.2,157 > 0	PASS
Satisfaction range	All scores within 1–4	PASS
Tenure logic	Years in role $\leq$ years at company	PASS
Attrition rate range	23.5% within realistic 15–35%	PASS
T-test significance	$p = 0.000003 < 0.05$ confirmed	PASS
Correlation direction	Overtime +ve, income −ve with attrition	PASS
Department coverage	All 7 departments present	PASS
Test	What is Checked	Result
Data shape	1,470 rows $\times$ 26 columns	PASS

KEY FINDINGS SUMMARY

Finding	Evidence
Overall attrition	23.5% — above healthy benchmark
Biggest driver	Overtime: leavers at 38% vs 17%
Salary impact	Income < Rs.3,000 leads to very high exit
Highest risk age	22–25 group — 38% attrition
Satisfaction effect	Score 1 → 46% leave, Score 4 → 11% leave
Worst department	Sales — 29.4% attrition
Best department	Marketing — 18.4% attrition
Training benefit	3–4 sessions/year → attrition drops to 18%